

Devixo Solutions AI/ML Internship

Task 01 – Python Programming, Data Analysis & Visualization

Dataset: Punjab Crime Statistics

Dataset Size: 6,885 rows × 6 columns

Tools: Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn

1. Introduction

This project was completed as part of the Devixo Solutions AI/ML Internship Task 01. The purpose of the task is to build practical skills in Python-based data analysis and visualization using a real-world dataset.

The selected dataset contains Punjab crime statistics across years, divisions, districts, crime types, crime counts, and population values. The analysis focuses on understanding the structure of the data, checking data quality, identifying patterns, calculating descriptive statistics, and communicating findings through visualizations.

2. Objective

The main objectives are to load and understand the dataset, identify and handle data-quality issues, perform exploratory data analysis, calculate descriptive statistics and correlations, create meaningful visualizations, and generate evidence-based insights.

3. Dataset Information

Item	Details
Dataset	Punjab Crime Statistics
Rows	6,885
Columns	6
Columns	Year, Division, District, CrimeType, CrimeCount, Population
Years present	2002–2015
Divisions	9
Districts	36
Crime types	15

Feature description: Year represents the reporting year. Division and District identify the geographic area.

CrimeType identifies the type/category of crime. CrimeCount represents the recorded count. Population represents the population associated with the geographic record.

4. Methodology

Dataset Loading → Data Understanding → Data Cleaning → Exploratory Data Analysis → Statistical Analysis → Data Visualization → Insights & Findings → Conclusion.

5. Data Loading & Understanding

The CSV dataset was loaded using Pandas. Initial inspection used `head()`, `tail()`, `sample()`, `info()`, `shape`, column names, data types, and missing-value checks. The dataset contains six columns: Year, Division, District, CrimeType, CrimeCount, and Population.

6. Data Cleaning

The dataset contains 0 missing values in total and 0 duplicate rows. Because no missing values or duplicate rows were found, no imputation or duplicate removal was required. The numerical columns were already stored as numeric data types, and the categorical fields were stored as text.

Data-quality checks should also include checking for negative crime counts/population values and reviewing the year range.

7. Exploratory Data Analysis

The exploratory analysis examined crime-type frequency, district-level totals, yearly totals, and the distribution of numerical variables. This helps identify which categories and locations contribute most to the recorded crime counts and how totals change over time.

8. Statistical Analysis

Descriptive statistics were calculated using mean, median, minimum, maximum, standard deviation, and variance. A correlation matrix was also calculated for Year, CrimeCount, and Population.

Statistic	CrimeCount	Population
Mean	1,332.47	2,613,793.76
Median	163.00	2,346,000.00
Minimum	0.00	17.70
Maximum	86,707.00	9,744,000.00
Standard deviation	4,542.09	1,601,153.29
Variance	20,630,593.05	2,563,691,854,239.32

Correlation between Population and CrimeCount is 0.302, indicating a positive but not strong linear relationship in the row-level data. Correlation between Year and CrimeCount is 0.032, indicating a very weak linear relationship at the row level.

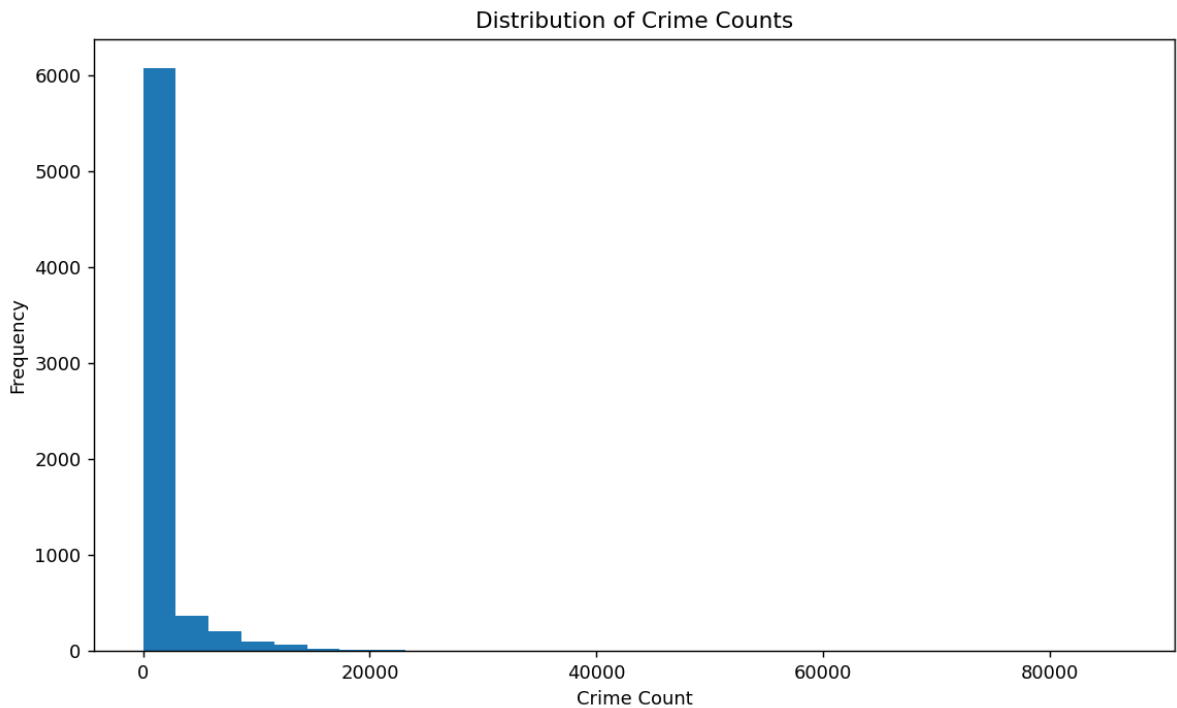
9. Major Findings

1. The dataset contains 6,885 records across 36 districts and 15 crime-type categories.
2. Lahore has the highest total recorded CrimeCount among districts in this dataset, with 1,757,905 recorded counts across its rows.
3. Faisalabad is the second-highest district by total recorded CrimeCount, with 736,488.
4. The yearly total is lowest in 2002 (495,624) and reaches its highest value in 2012 (790,012).
5. The yearly totals rise through the later part of the period and then decline slightly after the peak, showing that the overall trend is not a simple straight-line increase.
6. CrimeCount is positively correlated with Population, but the correlation is moderate rather than close to +1, so population alone does not explain all variation in recorded crime counts.
7. CrimeCount has substantial variation, which is visible in the histogram and box plot.
8. The box plot indicates that some records have substantially larger CrimeCount values than the typical record, making outlier analysis important.
9. The crime-type frequency chart shows that the dataset has a repeated structure across geographic/year records, with each crime category appearing many times.
10. Combining categorical, numerical, geographic, and time-based views provides a more complete understanding than relying on a single chart.

10. Data Visualizations

Visualization 01 – Histogram

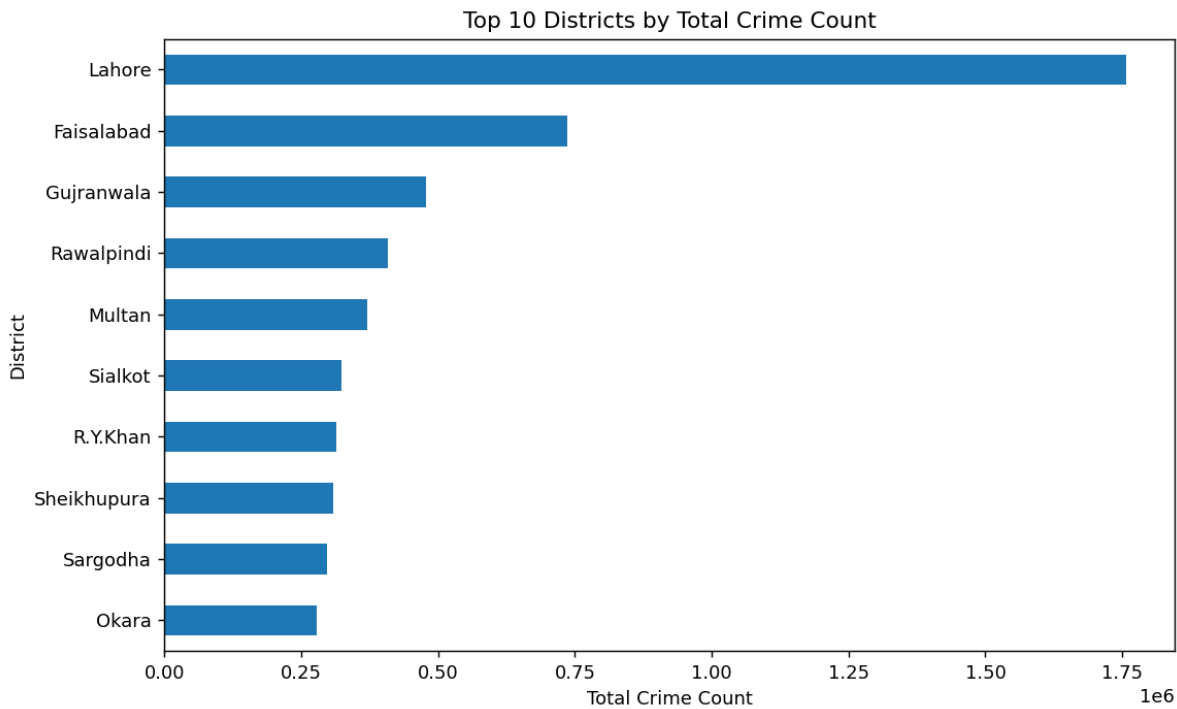
Shows the distribution of CrimeCount and helps identify concentration, spread, and unusually high values.



Interpretation: The figure above should be read together with the stated pattern and the underlying dataset values.

Visualization 02 – Bar Chart

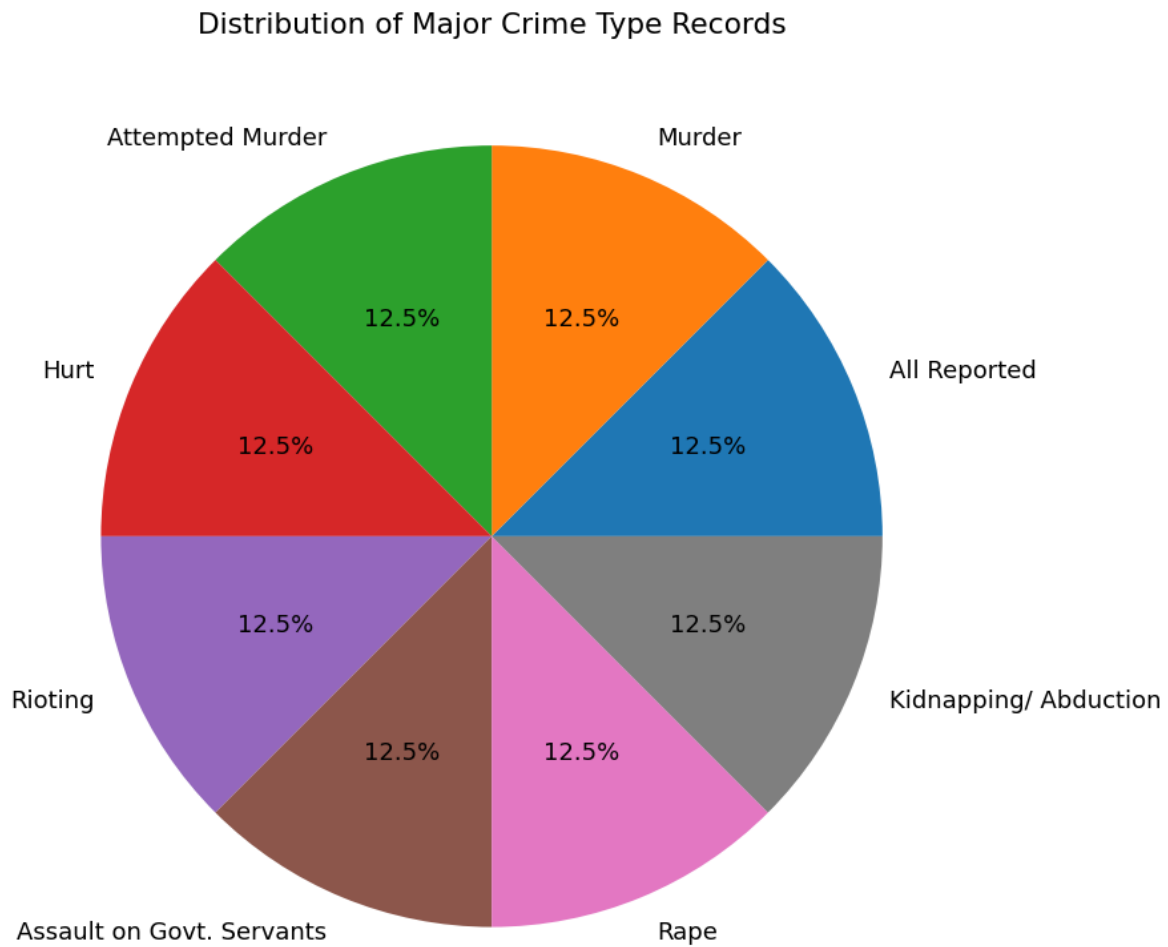
Compares total CrimeCount across the top ten districts and highlights districts with the largest recorded totals.



Interpretation: The figure above should be read together with the stated pattern and the underlying dataset values.

Visualization 03 – Pie Chart

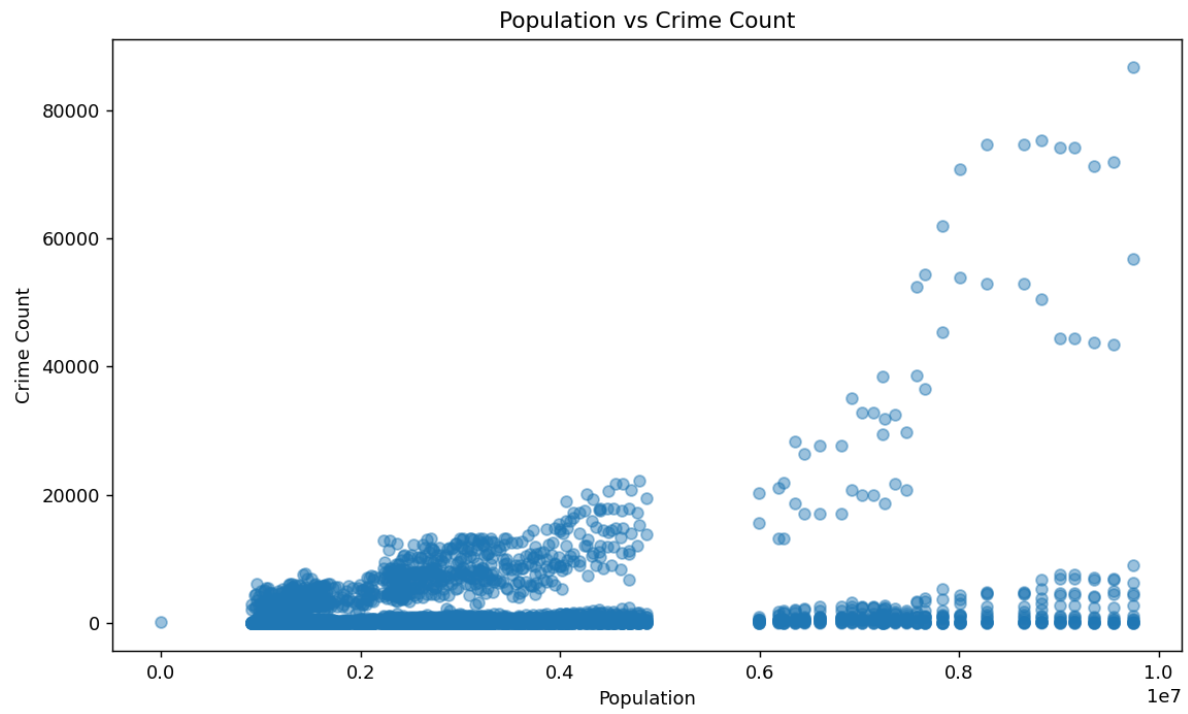
Shows the share of the most frequently occurring crime-type records in the dataset.



Interpretation: The figure above should be read together with the stated pattern and the underlying dataset values.

Visualization 04 – Scatter Plot

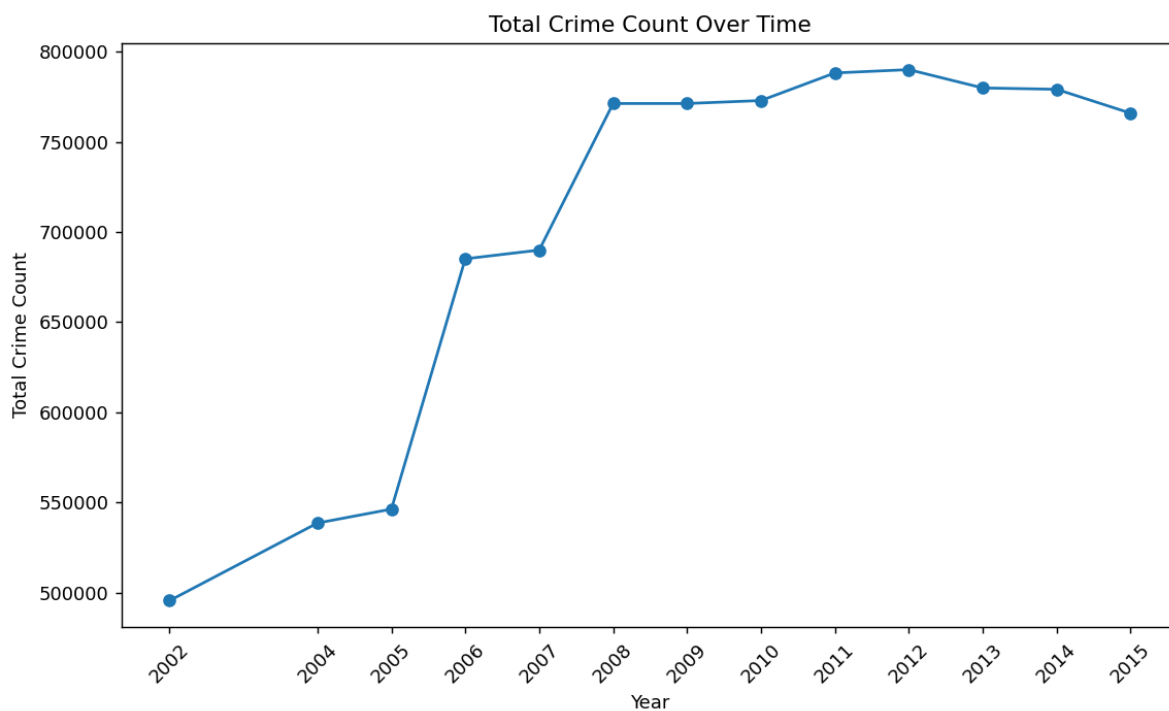
Shows the relationship between Population and CrimeCount; the points indicate substantial variation around the overall positive relationship.



Interpretation: The figure above should be read together with the stated pattern and the underlying dataset values.

Visualization 05 – Line Chart

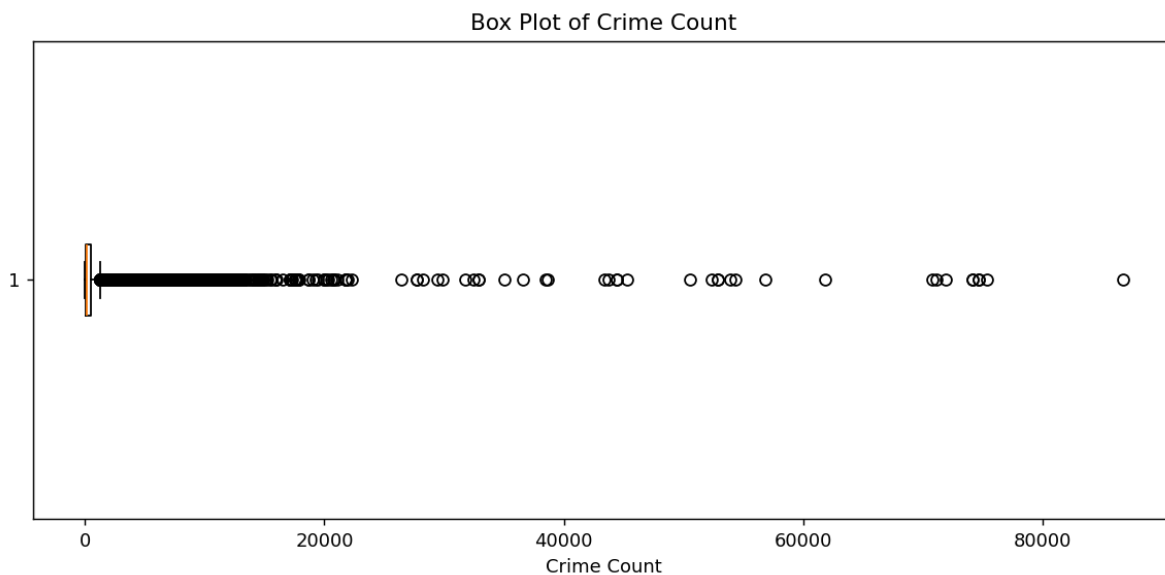
Shows how the total recorded CrimeCount changes across the years present in the dataset.



Interpretation: The figure above should be read together with the stated pattern and the underlying dataset values.

Visualization 06 – Box Plot

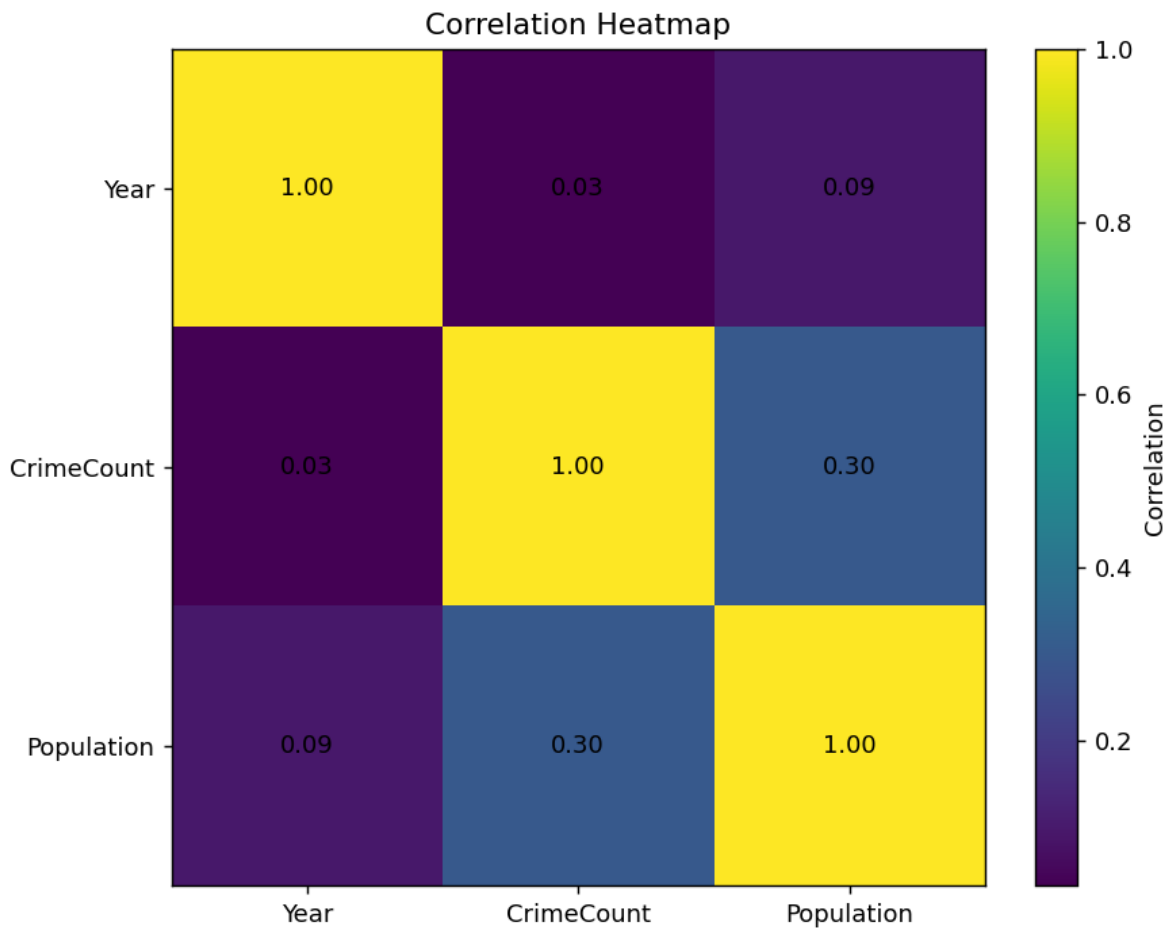
Summarizes the spread of CrimeCount and highlights unusually high observations.



Interpretation: The figure above should be read together with the stated pattern and the underlying dataset values.

Visualization 07 – Correlation Heatmap

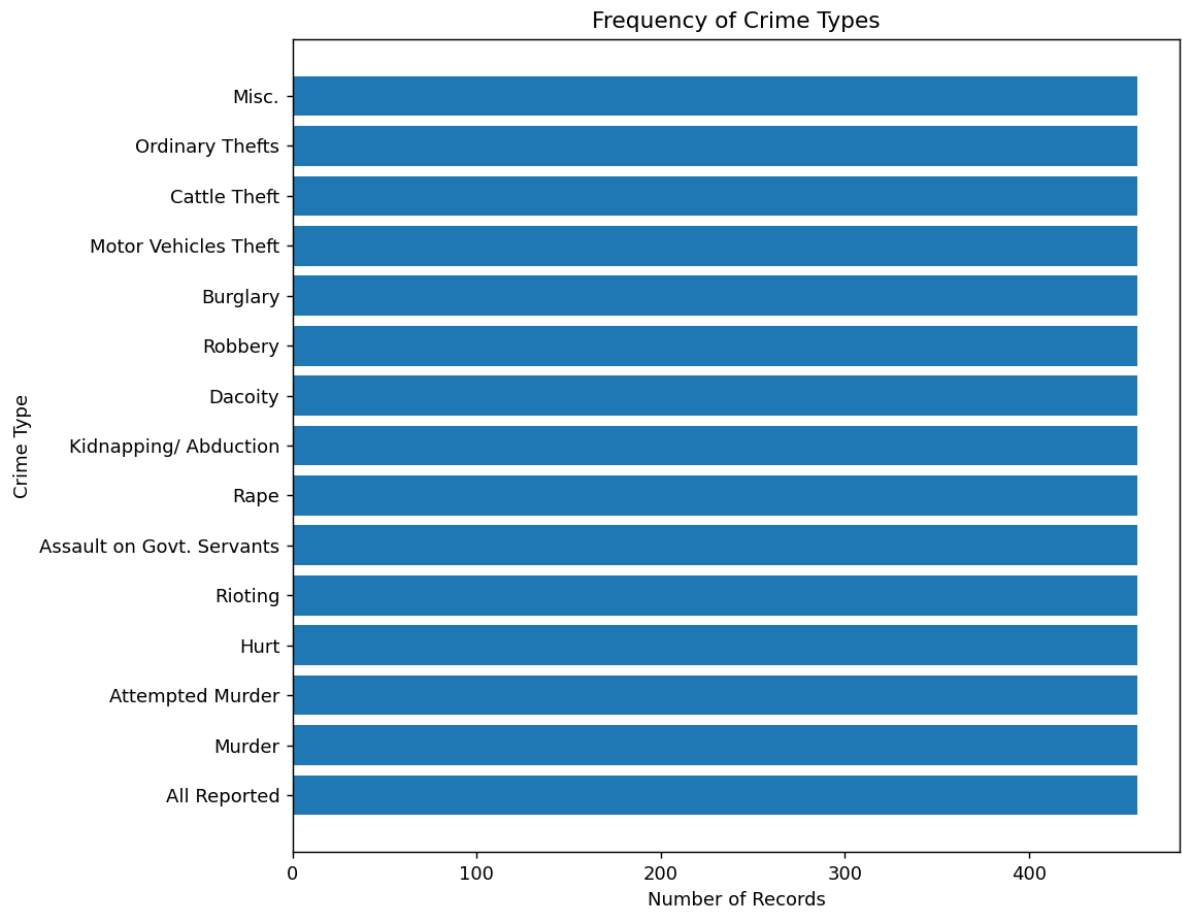
Displays correlations among Year, CrimeCount, and Population, making positive and weak relationships easy to compare.



Interpretation: The figure above should be read together with the stated pattern and the underlying dataset values.

Visualization 08 – Count Plot

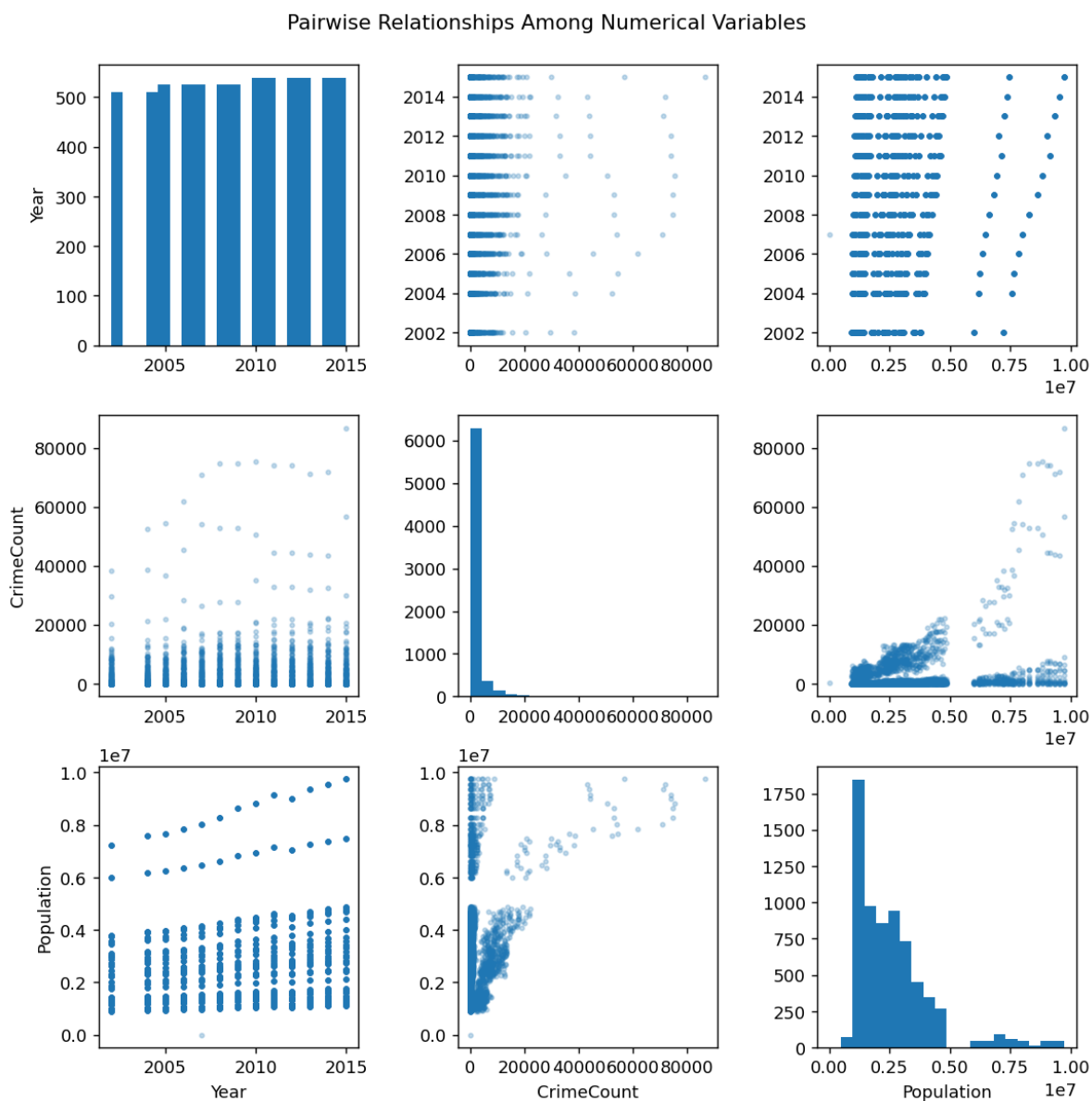
Shows how frequently each CrimeType appears as a record/category in the dataset.



Interpretation: The figure above should be read together with the stated pattern and the underlying dataset values.

Visualization 09 – Pair Plot

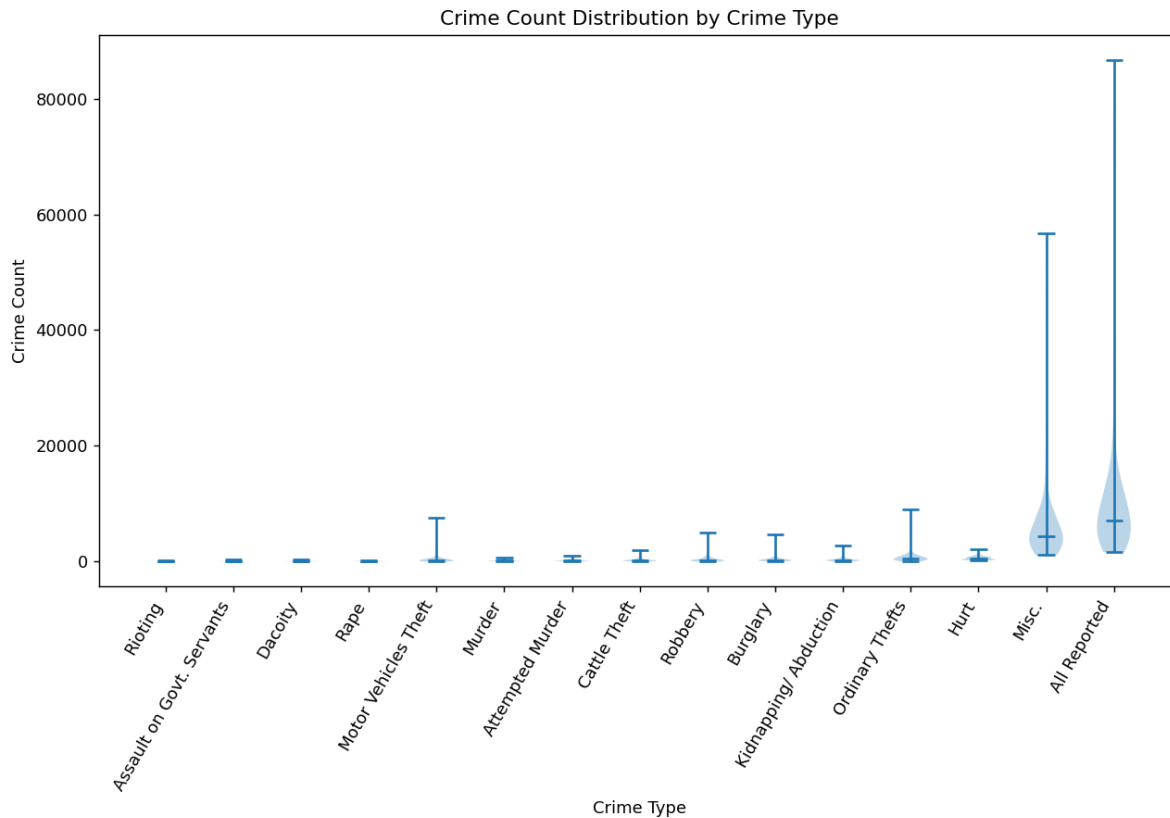
Provides a compact view of pairwise relationships and distributions among the numerical variables.



Interpretation: The figure above should be read together with the stated pattern and the underlying dataset values.

Visualization 10 – Violin Plot

Shows the distribution and spread of CrimeCount for each crime type, helping compare variation between categories.



Interpretation: The figure above should be read together with the stated pattern and the underlying dataset values.

11. Screenshots / Evidence

For the final submission, notebook screenshots should be embedded in this report as evidence of the actual work.

```
#Load data set
df = pd.read_csv("../dataset/punjab-crime-stats.csv")
#list five rows
print(df.head())
```

	Year	Division	District	CrimeType	CrimeCount	Population
0	2002	Bahawalpur	Bahawalpur	All Reported	8440	2743000.0
1	2002	Bahawalpur	Bahawalpur	Murder	105	2743000.0
2	2002	Bahawalpur	Bahawalpur	Attempted Murder	226	2743000.0
3	2002	Bahawalpur	Bahawalpur	Hurt	700	2743000.0
4	2002	Bahawalpur	Bahawalpur	Rioting	12	2743000.0

```
#last 5 rows
print(df.tail())
```

	Year	Division	District	CrimeType	CrimeCount	Population
6880	2015	Sargodha	Mianwali	Burglary	68	1427000.0
6881	2015	Sargodha	Mianwali	Motor Vehicles Theft	53	1427000.0
6882	2015	Sargodha	Mianwali	Cattle Theft	39	1427000.0
6883	2015	Sargodha	Mianwali	Ordinary Thefts	127	1427000.0
6884	2015	Sargodha	Mianwali	Misc.	3610	1427000.0

```
#info
print(df.info())
```

```
<class 'pandas.DataFrame'>
RangeIndex: 6885 entries, 0 to 6884
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Year            6885 non-null   int64
1   Division        6885 non-null   str
2   District        6885 non-null   str
3   CrimeType       6885 non-null   str
4   CrimeCount      6885 non-null   int64
5   Population      6885 non-null   float64
dtypes: float64(1), int64(2), str(3)
memory usage: 322.9 KB
None
```

```
#Crime
crime_type_count=df["CrimeType"].value_counts()
print(crime_type_count)
```

```
[ ]
```

```
...   CrimeType
All Reported          459
Murder                459
Attempted Murder     459
Hurt                  459
Rioting               459
Assault on Govt. Servants 459
Rape                  459
Kidnapping/ Abduction 459
Dacoity               459
Robbery               459
Burglary              459
Motor Vehicles Theft  459
Cattle Theft          459
Ordinary Thefts       459
Misc.                 459
Name: count, dtype: int64
```

```
#Mode
print("Mod Crimee Count:", df["CrimeCount"].mode())
print("Mode Popultion:", df["Population"].mode())
```

```
Mod Crimee Count: 0      0
Name: CrimeCount, dtype: int64
Mode Popultion: 0      1196000.0
1      1203000.0
2      1211000.0
3      1243000.0
4      1267000.0
5      1304000.0
6      1309000.0
7      1353000.0
8      1368000.0
9      1390000.0
10     1408000.0
11     1412000.0
12     1430000.0
13     1486000.0
14     1514000.0
15     1529000.0
16     1631000.0
17     2423000.0
18     2430000.0
19     2454000.0
20     2618000.0
21     3994000.0
```

12. Conclusion

This project provided practical experience in Python-based data analysis using Pandas, NumPy, Matplotlib, and Seaborn. The Punjab Crime Statistics dataset was loaded and inspected, and its structure, data types, missing values, duplicates, and numerical distributions were examined. Because the uploaded dataset contained no missing values or duplicate rows, no imputation or duplicate removal was necessary.

The exploratory and statistical analysis showed meaningful differences across districts, crime categories, and years. Lahore had the highest district-level total in the selected data, yearly totals reached their maximum in 2012, and Population had a positive but moderate relationship with CrimeCount. The ten visualizations help communicate these patterns and demonstrate how data analysis can support evidence-based interpretation of real-world datasets.

13. GitHub Repository Structure

```
Task_01_AI_ML/
├── dataset/Punjab_Crime_Stats.csv
├── notebook/Task_01_Data_Analysis.ipynb
├── images/visualization_01.png ... visualization_10.png
└── report/Task_01_Report.pdf
```

GitHub Repository Link: https://github.com/rehanfayyaz014/Task_01_AI_ML